

KRALJEVO, Serbia

WMO: 133760

Lat: 43.700N

Lon: 20.700E

Elev: 217

StdP: 98.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.1	-13.6	1.2	-9.8	-11.0	1.5	-7.7	9.0	5.8	7.9	2.9	1.2	320	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	34.3	20.9	32.4	20.9	30.7	20.5	23.1	30.2	22.1	29.3	21.2	28.3	2.4	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	16.2	26.1	20.0	15.0	24.8	19.0	14.1	24.1	69.7	29.9	65.7	29.0	62.5	28.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.0	5.1	-16.3	36.8	4.8	2.2	-19.7	38.4	-22.5	39.7	-25.2	41.0	-28.6	42.6		
			-16.4	24.6	4.6	1.2	-19.7	25.5	-22.4	26.2	-24.9	26.8	-28.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	0.5	2.9	7.0	12.1	16.5	20.3	22.2	17.3	12.0	7.2	1.7	
	DBStd	8.61	4.89	5.06	4.52	3.75	3.45	3.39	3.00	2.94	3.63	4.06	4.47	
	HDD10.0	1028	294	202	114	22	1	0	0	1	27	108	259	
	HDD18.3	2736	552	432	352	189	77	19	4	60	197	334	515	
	CDD10.0	1714	1	4	21	84	202	309	378	221	90	24	3	
	CDD18.3	379	0	0	0	1	19	78	124	124	31	2	1	0
	CDH23.3	4322	0	0	2	52	281	858	1370	1335	379	44	1	0
	CDH26.7	1739	0	0	0	9	59	323	603	608	132	5	0	0
Wind	WSAvg	1.7	1.7	1.9	2.1	2.0	1.6	1.6	1.5	1.4	1.5	1.5	1.8	1.7
Precipitation	PrecAvg	787	48	49	63	69	87	87	72	60	72	64	56	60
	PrecMax	1176	108	76	158	177	195	192	192	143	173	140	111	123
	PrecMin	539	11	14	11	28	39	28	14	0	13	2	2	8
	PrecStd	140	28	18	36	34	38	42	43	37	40	35	28	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	18.8	23.0	28.0	30.2	34.1	37.3	37.9	33.8	27.0	21.9	16.8
		MCWB	9.9	11.1	12.7	16.4	19.0	21.8	20.9	20.7	19.6	17.2	14.4	11.5
	2%	DB	11.9	15.9	20.4	24.3	28.2	32.1	34.3	34.8	30.1	24.2	19.2	13.3
		MCWB	8.0	10.0	11.6	14.8	18.4	21.6	21.2	20.7	18.8	16.5	13.5	9.3
	5%	DB	9.8	13.2	18.1	22.0	26.7	30.2	32.1	32.3	27.2	22.0	17.0	10.6
		MCWB	6.9	8.4	10.5	13.5	17.9	20.9	21.2	20.5	18.0	15.6	12.2	8.0
	10%	DB	7.4	10.8	15.6	19.9	24.6	28.3	30.1	30.1	24.9	19.7	14.8	8.0
		MCWB	5.4	7.3	9.5	12.7	17.0	20.0	20.8	20.3	17.4	14.8	11.1	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	12.0	13.5	17.3	20.6	24.5	24.6	24.2	21.2	18.7	15.4	11.9
		MCDB	14.4	17.3	21.0	25.5	27.3	30.5	29.5	30.8	28.3	25.2	20.3	16.0
	2%	WB	8.5	10.3	12.1	15.5	19.4	22.8	23.1	22.7	19.8	17.2	13.8	9.8
		MCDB	11.5	15.2	18.7	22.8	26.2	29.6	30.2	30.1	27.1	22.2	18.2	12.7
	5%	WB	7.0	8.9	11.0	14.3	18.5	21.6	22.1	21.7	18.9	16.1	12.6	8.1
		MCDB	9.3	12.6	17.0	20.7	25.0	28.3	29.4	29.0	25.6	20.7	16.3	10.4
	10%	WB	5.5	7.6	9.8	13.2	17.5	20.6	21.2	20.9	18.0	15.0	11.4	6.4
		MCDB	7.3	10.6	15.2	18.5	23.5	27.1	28.5	27.9	23.7	19.4	14.5	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.0	10.7	11.7	12.1	12.1	13.0	13.5	12.1	11.4	9.4	7.0
		MCDBR	11.4	14.5	17.2	17.1	16.3	16.1	17.4	18.6	17.4	16.0	13.3	11.3
		MCWBR	7.8	9.0	9.3	8.5	7.5	6.7	6.4	6.5	7.3	8.6	8.0	8.0
	5% WB	MCDBR	10.8	13.1	15.2	14.6	14.4	14.5	14.8	14.7	14.7	13.9	12.1	10.4
		MCWBR	7.7	8.3	8.5	7.6	7.2	6.7	6.5	6.3	6.7	7.9	7.5	7.6
Clear-Sky Solar Irradiance	taub	0.336	0.364	0.403	0.463	0.455	0.446	0.447	0.459	0.436	0.394	0.354	0.333	
	taud	2.341	2.239	2.193	2.054	2.137	2.208	2.206	2.169	2.205	2.315	2.372	2.366	
	Ebn at Noon	775	812	826	803	822	830	824	798	784	771	747	741	
	Edh at Noon	82	107	128	159	150	141	139	140	125	99	80	73	
All-Sky Solar Radiation	RadAvg	1.39	2.21	3.34	4.34	5.31	6.00	6.18	5.59	3.91	2.72	1.72	1.21	
	RadStd	0.21	0.27	0.41	0.52	0.54	0.59	0.48	0.53	0.40	0.34	0.19	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	+0.65	N/A	N/A	N/A	+0.39	N/A	N/A	-149	+105	N/A

Nomenclature: See separate page